

Symbol Matching

Goal

Build a workflow that allows our team to identify a symbol once on a drawing page, then automatically find and extract matching instances across the relevant drawing set.

User Flow

A user can draw a box around a symbol on a page and trigger an action such as **"Identify"** or **"Extract."**

After selecting the symbol, the user sees a modal with extraction options:

1. **This page only**

Run symbol matching only on the current page.

2. **All pages with a similar page name / plan type**

Run symbol matching across pages that appear to be the same drawing type.

For example, if the current page is **First Floor Power Plan**, the system should also run against pages like:

- Second Floor Power Plan
- Third Floor Power Plan
- Basement Power Plan

3. **All pages within the same pageType**

Run symbol matching across all pages with the same broader pageType.

For example:

- Architectural pages
- Electrical pages
- Mechanical pages
- Plumbing pages

System Behavior

Once the user selects the desired scope, the system should:

1. Use the boxed region as the reference symbol.
2. Search the selected pages for visually matching instances of that symbol.
3. Generate `drawingItems` in our database for each matched instance.
4. Generate associated `captures` in our database for each matched instance.
5. Preserve page-level metadata for each result, including available fields such as:
 - Page name, e.g. `First Floor Power Plan`
 - Sheet reference, e.g. `E201`
 - `pageType`
 - Bounding box / capture coordinates
 - Source page ID

Available Inputs

We already have access to useful page metadata, including:

- Page names, such as `First Floor Power Plan`
- Sheet references, such as `E201`
- `pageType` values
- Page grouping / classification data, where available

This metadata can be used to determine which pages should be included in the matching run.

Extension / Bonus Scope

Build a related flow for detecting non-symbol visual patterns, markings, or shaded regions that our team currently has to identify manually.

Examples include:

- Reflected Ceiling Plan shading or hatch patterns

- Wall type patterns, such as CMU vs. metal framed partitions
- Other repeated visual markings that are not clean, isolated symbols

The goal of this extension would be to let users identify a visual condition once, then detect similar conditions across relevant pages in the same way as symbol matching.